



POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Data Science in Chemical Engineering

Neural Surrogate Modeling of a Reaction–Separation Process

A.Y. 2025–2026

Context and motivation

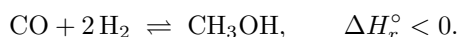
Rigorous process simulators (Aspen Plus, DWSIM, gPROMS, Cantera-based chains) are the workhorse of chemical process design, but they are computationally expensive and non-differentiable. This becomes a bottleneck whenever a process model must be evaluated thousands of times: multi-start optimization, uncertainty quantification, real-time optimization, operator training simulators, and digital twins all suffer from the cost of the underlying rigorous model.

A standard remedy is *surrogate modeling*: a cheap, differentiable approximation of the rigorous simulator is trained on a dataset of input–output pairs and then used in place of the simulator for the downstream task. Among the many surrogate families (polynomial chaos, Gaussian processes, random forests, radial basis functions), **feed-forward neural networks** have become a default choice because they scale well with data, are trivially differentiable, and integrate with modern optimization frameworks.

In this project you will build, train, validate, and apply a neural surrogate for a simplified **methanol synthesis loop**. The emphasis is on the **complete data science workflow**, not on any single step in isolation.

Process description

The process is a once-through, simplified version of the low-pressure methanol synthesis loop. Fresh syngas (stream 1, CO and H₂) is preheated in exchanger HX1 with duty Q_1 and mixed with a secondary H₂-rich make-up stream (stream 2) at the reactor inlet. The reactor R operates at pressure P_R and hosts the main methanol synthesis reaction over a Cu/ZnO/Al₂O₃-type catalyst:



The reactor effluent enters a flash drum operating at $(T_{\text{flash}}, P_{\text{flash}})$: unconverted syngas leaves as vapor at the top (stream 3), while the condensed crude methanol (with a small amount of water as a trace by-product) leaves as liquid at the bottom. The liquid is further cooled in exchanger HX2 with duty Q_2 and leaves the battery limits as stream 4 (see Figure 1).

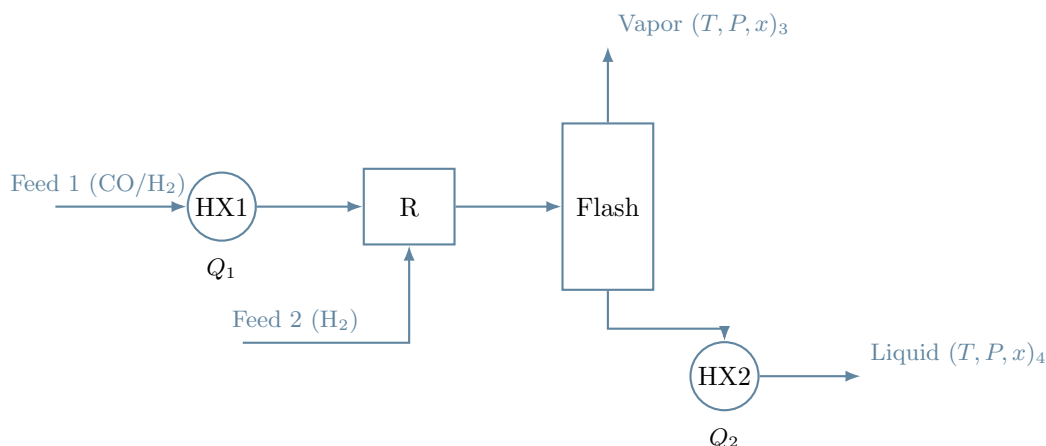


Figure 1: Simplified methanol synthesis loop: two feeds, one equilibrium reactor, one flash separator, two heat exchangers.

The rigorous simulation of this process is based on mass and energy conservation, van’t Hoff chemical equilibrium for the reactor, Antoine vapor pressures and a Rachford–Rice flash for the separator. The four species involved are CO, H₂, CH₃OH and H₂O. Compared to a full industrial plant, the model is a deliberate simplification: the water-gas-shift reaction and the gas recycle are not considered, and CO₂ is not part of the feed.

The dataset required to complete the project is provided as a zip archive, `dataset.zip`, which contains:

- `train.csv` — a training set generated via Latin Hypercube Sampling over the input domain;
- `test.csv` — an independent test set generated via random sampling;
- `inputs_schema.md` — names, units, physical meaning and bounds of every input variable;
- `outputs_schema.md` — names, units and physical meaning of every output variable;
- `README.md` — instructions and conventions.

For the purposes of building your surrogate, the only information you should rely on is what is contained in the provided datasets: the process will be treated as a **black box**. This mirrors the real industrial practice of building surrogates for proprietary or licensed simulators.

Tasks

Task 1. Exploratory data analysis. Inspect the dataset. Produce marginal and joint distributions of inputs and outputs; identify correlations, nonlinear couplings, and any output that is nearly constant or nearly linear in the inputs. Flag regions of input space where the process model produces degenerate outputs (single-phase flash, zero conversion, etc.) and discuss how these observations inform your subsequent modeling choices.

Task 2. Baseline models. Fit at least two baseline regressors: (i) multivariate **linear regression** and (ii) one nonlinear baseline of your choice (polynomial regression, random forest, or Gaussian process on a small subset). Report their performance on the test set. These baselines define the bar the neural surrogate must clear.

Task 3. Neural surrogate. Train a feed-forward neural network that maps process inputs to process outputs. Your report must document: input/output scaling, train/validation split, architecture search (at least three architectures compared), loss function, optimizer, learning rate schedule, regularization, and early stopping. Report final performance on the held-out test set using at least **MAE, RMSE, and R^2 per output variable**, produce parity plots, and briefly discuss in which regions of the input space the residuals are largest and why.

Task 4. Application: constrained optimization. Use your trained surrogate to find the operating conditions (Q_1 , Q_2 , P_R , T_{flash} , and any other decision variables you consider appropriate) that **maximize the methanol recovery in stream 4** subject to a constraint on total energy duty

$|Q_1| + |Q_2| \leq Q_{\max}$ with $Q_{\max} = 1.7$ MW, and on the methanol purity of stream 4. You may use any gradient-based or gradient-free optimizer (`scipy.optimize`, PyTorch autodiff, etc.). Report the optimal operating point predicted by your surrogate. The instructor will evaluate your optimum using the rigorous simulator and the agreement between surrogate prediction and simulator ground truth will be part of your final score.

Deliverables

1. A written report structured along the four tasks above.
2. A reproducible code repository (a single Git repo or a zipped folder) containing:
 - all scripts and notebooks used to train the models and run the optimization;
 - a Readme file explaining how to reproduce your results from scratch.